6.5 Rotation of Conics

Name _____ Period ____ Date _____

Pre-Calculus 6.5 · Student sheet

I can…

- find the angle of rotation that eliminates the xy-term.
- use the rotation formulas to rewrite an equation.
- classify a conic using the discriminant.

Vocabulary · fill in the missing word

Rotation of axes — _____ the x- and y-axes through an angle θ to get new x′- and y′-axes

xy-term — the term Bxy in $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$; it shows the conic is _____

Discriminant of a conic — _____

Part 1 · A tilted hyperbola

Graph $xy = 1$ and $x^2 - y^2 = 2$. Then add a slider t to rotate the second graph.

Graph	Shape	Angle t that makes them match
$xy = 1$		
$x^2 - y^2 = 2$ (rotated)		

Part 2 · Notes and examples (fill in each note, then work the example)

Angle of rotation: to eliminate the xy-term, rotate by θ with $\cot 2\theta =$ _____.

Example 1

Find the angle of rotation that eliminates the xy-term in $xy - 1 = 0$.

Rotation formulas: $x = x'\cos\theta - y'\sin\theta$, $y =$ _____

Example 2

Rotate the axes by 45° to rewrite $xy - 1 = 0$ in standard form.

Example 3

Find the angle of rotation for $7x^2 - 6\sqrt{3}\,xy + 13y^2 - 16 = 0$.

Discriminant: $B^2 - 4AC < 0$ ellipse or circle, $= 0$ parabola, _____ 0 hyperbola.

Example 4

Classify each conic.

a. $4xy - 9 = 0$

b. $2x^2 - 3xy + 2y^2 + 2x = 0$

c. $x^2 - 6xy + 9y^2 - 2y + 1 = 0$

Example 5

The axes are rotated 60°. Find the $x'y'$-coordinates of the point $(1, \sqrt{3})$.

Part 3 · Partner practice

Partner 1

Find the angle of rotation for $x^2 + xy + y^2 = 10$.

Partner 2

Classify $3x^2 + 8xy + 4y^2 - 7 = 0$.